

Catherine Zhang

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EDUCATION

University of Washington

Seattle, WA

Bachelor of Science in Computer Science, GPA: 3.66/4.0

Sep. 2024 – June 2028 (Expected)

- Coursework: Artificial Intelligence, Computer Graphics, Computer Vision, Data Structures and Algorithms, Hardware/Software Interface, Computational Probability

EXPERIENCE

Software Engineering Intern

June 2026 – September 2026

Aggro Crab Games | C#, Unity, Blender, Shader Graph

Seattle, WA

- Sole programming intern on an unannounced title. Designed a centralized entity manager that staggers visibility checks across coroutine cycles, amortizing per-entity work across frames and raising framerate by **25%**.
- Built a custom **Unity** editor tool that procedurally places prefabs along a spline, sampling at configurable density with surface-snapping raycasts and gizmo previews, replacing manual placement.
- Animated environmental assets through **Shader Graph** vertex-offset shaders, **Blender** shape keys, and rigged Unity Animator clips, and added **C#** interaction logic and custom particle effects.
- Playtested and reviewed builds for a major **PEAK** update (**10M+** copies sold), identifying and reporting gameplay bugs and giving feedback that informed design decisions ahead of release.

Undergraduate Research Assistant

August 2025 – December 2025

The Machine Agency at UW | JavaScript, G-Code, 3D Printing

Seattle, WA

- Drove **G-code** machine control directly through **p5.fab**, stress-testing serial communication and specifying toolpath streaming fixes compatible with **4+** industry-standard printers.
- Scripted custom toolpaths that bypass conventional CAD/CAM, tuning low-level machine parameters to produce non-planar tactile graphics for visually impaired learners.

Firmware Developer

Oct. 2024 – June 2025

Washington Superbike | C++, Arduino, Raspberry Pi Pico, Teensy, CAN Bus

Seattle, WA

- Designed a Hardware-in-the-Loop (HIL) test rig in **C++** across Arduino, Raspberry Pi Pico, and Teensy to validate electric motorcycle motor controllers before hardware deployment.
- Built a **CAN bus** fault injection harness simulating battery overheating and precharge circuit failures, cutting undetected fault conditions by **~70%** before physical testing.

Software Engineering Intern

Aug. 2023 – July 2024

Fair Worlds | JavaScript, HTML, 8th Wall.js, Adobe Aero

Seattle, WA

- Shipped geospatial web AR experiences in 8th Wall.js for the Seattle Art Tour: 3D-scanned **50+** public installations, animated assets in Adobe Aero, and implemented image recognition for AR navigation.

PROJECTS

Terrain-simulator | C++, WebAssembly, React, Node.js, CSS

September 2026

- Compiled a **C++** hydraulic erosion simulator to **WebAssembly** via **Emscripten**, driven from a **React/Vite** interface exposing **10** live parameters, with terrain seeded by **fractal Brownian motion** over value noise.
- Cut erosion runtime by **35%** across **4** profiler-guided optimizations, including precomputed lookup tables and reusing interpolated state between simulation steps.

The Mirrors | Lead SWE | Unity, C#, Adobe Suite

December 2025 – March 2026

- Co-developed a horror-mystery game in **Unity** around an enemy visible only in mirror reflections, requiring reflected geometry consistent with the player's real-time viewpoint.
- Engineered a custom **ray tracing** system in **C#** to compute mirror reflections, correcting perspective distortion from off-axis viewing angles and non-coplanar surfaces.

TECHNICAL SKILLS

Languages: Java, C#, C/C++, JavaScript, Python, HTML/CSS, G-code, Assembly

Frameworks & Tools: Git, React, Node.js, WebAssembly, Vite, Unity, Shader Graph, HLSL, Blender, Emscripten, Arduino, Adobe Suite